

Algebra 1
Unit 13
Lesson 9 Practice

Name _____
Date _____
Period _____

To make a 14% acid solution, a chemist mixes 4 parts of 2% acid solution with an 18% acid solution. The expression $4(0.02x) + 0.18y$ represents the mixture, where x is the number of ounces of 2% solution and y is the number of ounces of the 18% solution.

Explain what each quantity means.

1. $(0.02x)$

2. $4(0.02x)$

3. $0.18y$

Since 2018, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1,675,000(1.064)^x$.

4. What does the x variable represent?

5. What does the y variable represent?

6. What does the quantity 1,675,000 represent?

7. What is the constant percent rate of change?

The equation $y = -4(x - 2)^2 + 20$ models the height, y , in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 2$

9. 20

Michael is buying coffee for \$5.75 per bag. He has a coupon for three free bags of coffee. He pays \$25.50 for a birthday cake. The expression $5.75(x - 3) + 25.50$ can be used to represent the overall cost.

10. What does the quantity $5.75(x - 3)$ represent?